Year 5 Science Mission: Seeds of Science

National Science Week 2026

What could you discover if you investigated like a real scientist?

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What is National Science Week?

🌏 A National Celebration

Scientists, students and communities across Australia explore ideas together every August.

🌱 This Year's Theme

"Seeds of Science: nurturing science for all" — a seed starts small but grows into something amazing, just like your scientific ideas.

💡 Why It Matters

Every invention, medicine and technology began with someone asking a question and investigating the answer. Your curiosity could lead to the next big discovery.

🌱 **Think about it:** What do you think scientists do every single day?

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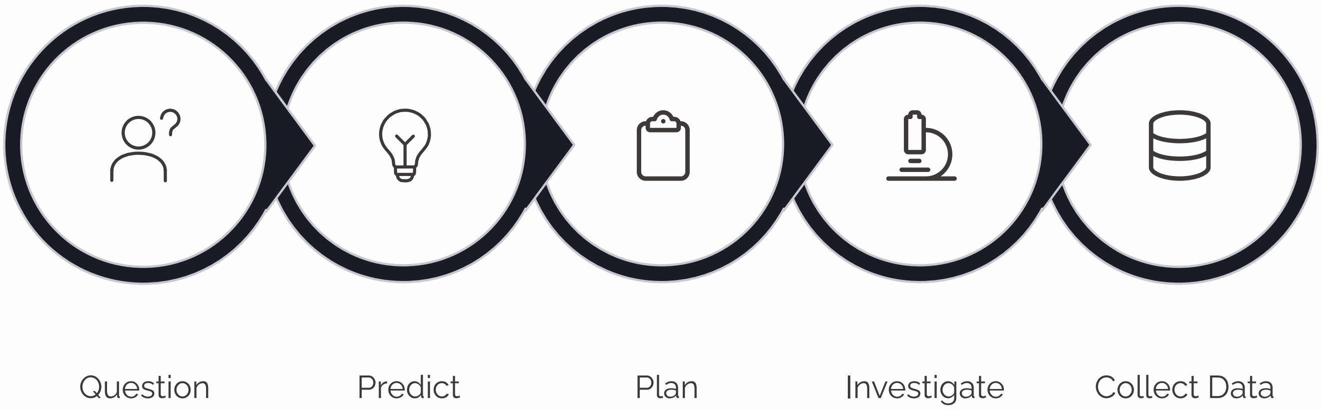
Think Like a Scientist

Real scientists follow a process to investigate and understand the world. Every great discovery starts with these seven steps:



Why Does Evidence Matter?

Evidence is the information you collect. It's what proves your ideas are correct — or shows you need to think differently. **Scientists trust evidence, not just guesses.**

🔍 **Prediction Challenge:** If you drop a ball and a feather at the same time in a room with air, which hits the ground first? What would happen in a vacuum with no air?

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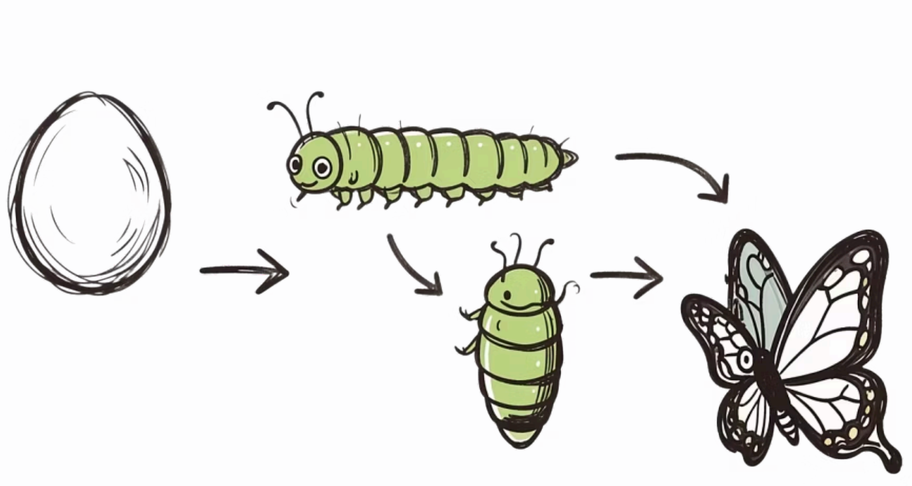
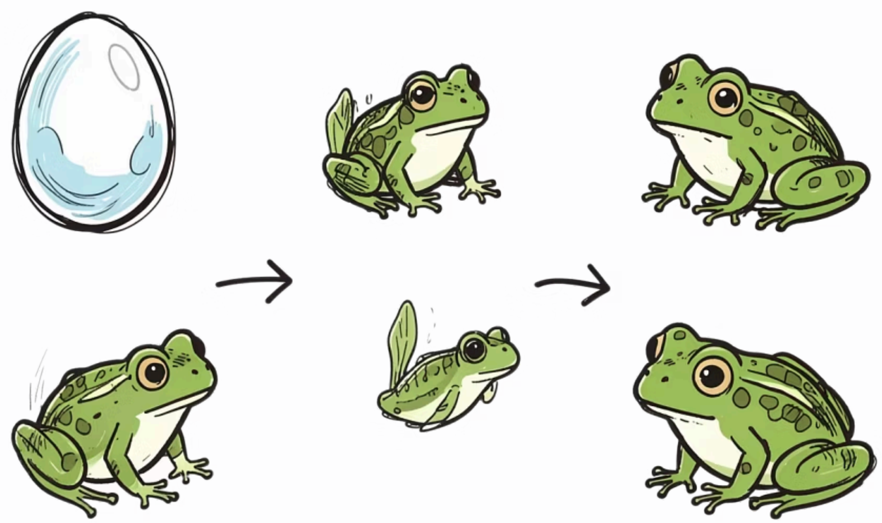
Life Cycles & Change

All living things go through stages of change called a **life cycle**. Compare these two remarkable transformations:





🦋 Butterfly Life Cycle

Egg → Caterpillar → Chrysalis → Butterfly

A **complete change** — the caterpillar looks nothing like the adult butterfly. Each stage has a different job: eat, transform, reproduce.

🐸 Frog Life Cycle

Egg → Tadpole → Froglet → Frog

A **gradual change** — the tadpole slowly grows legs and loses its tail. You can see it changing step by step.

🤔 **Reasoning Question:** Why do you think a caterpillar needs to change so much before it becomes a butterfly? What advantages does each stage give it?

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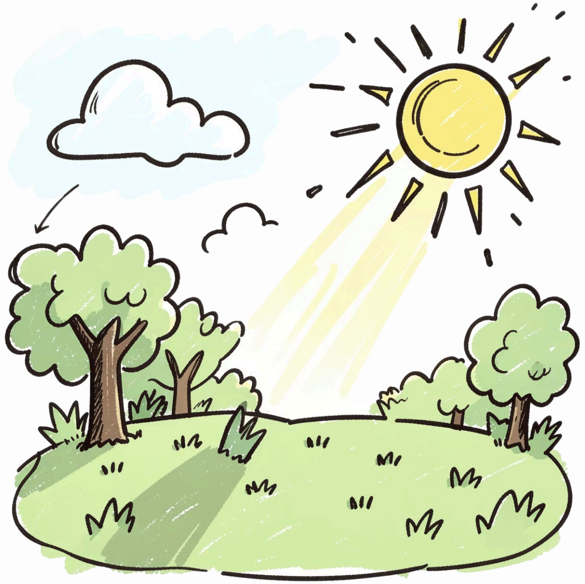

Day, Night & Shadows

Earth **rotates (spins)** on its axis. This rotation creates day and night — and it also changes the shadows around us throughout the day.

How Shadows Change:

📍 **Predict:** If you measure your shadow at 9 am, 12 pm and 3 pm, which will be shortest? Why?

🌅 Morning

Sun is low in the east — shadows are **long**

☀️ Midday

Sun is high in the sky — shadows are **short**

🌇 Afternoon

Sun is low in the west — shadows are **long again**

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Heat & Temperature

How Heat Moves

Heat is energy that moves from hot objects to cooler objects. **Temperature** measures how hot something is. Heat always travels from **hot to cold**.

Everyday Examples:

- A **metal spoon** in hot soup gets hot quickly → conductor
- A **wooden spoon** stays cool longer → insulator
- A **blanket** keeps you warm by trapping heat → insulator

🔬 **Investigation Scenario:** Which would best keep a hot drink warm – aluminium foil, bubble wrap, or cotton wool? Predict and explain why. What evidence would prove you right?

⚡ Conductors	🛡️ Insulators
Let heat pass through easily	Stop heat from passing through
Examples: metal, glass	Examples: foam, wool, plastic

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Living Things & Environments

Every living thing depends on its environment to survive. The environment provides food, water, shelter and the right conditions for life.

Desert Plants

Thick, waxy leaves save water. Spines instead of leaves protect precious moisture stores.

Polar Bears

Thick fur and a layer of blubber trap heat and insulate against freezing Arctic temperatures.

Fish

Gills extract oxygen from water, allowing fish to breathe in an environment where lungs would fail.

Rainforest Frogs

Moist, permeable skin works perfectly in humid rainforests — but would dry out fast in heat.

🌍 **Challenge Question:** What might happen to a rainforest frog if its home became dry and hot? Why would it struggle? What adaptations would help it?

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Mini Investigation: Which Material is the Best Insulator?

🔬 Question

Which material best keeps heat in — aluminium foil, bubble wrap, or cotton wool?

✏️ Prediction

I predict _________ will be the best insulator because _________

Variables

⚠️ ⚠️ **Safety:** Hot water can cause burns. Always let the teacher handle hot water. Wear safety glasses. Be careful with thermometers.

📋 Procedure

1. Wrap each cup with a different material (or leave one unwrapped as a control)
2. Pour the same amount of hot water into each cup
3. Record the starting temperature
4. Measure the temperature every 2 minutes for 10 minutes
5. Record all results in your data table

📊 Data Table

Time (min)	Foil (°C)	Bubble Wrap (°C)	Cotton Wool (°C)
0			
2			
4			
6			
8			
10			

Which cup lost the **least** heat? That material is the best insulator. The **smallest temperature drop = best insulation**. Did your results match your prediction?

Independent

The material being tested

Dependent

How much the temperature drops

Control

Same water amount, container, starting temperature, time period, room temperature

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Science Week Challenge Questions

Answer these questions using **evidence and reasoning** — not just guesses!

1	**Life Cycles** A butterfly caterpillar eats leaves all day and grows very quickly. Why does it need to eat so much? What is it preparing for?
2	**Shadows** A student's shadow is 2 metres at 8 am but only 0.5 metres at 12 pm. Explain why the shadow got shorter. What is causing this change?
3	**Sleeping Bag Design** A camping company needs a sleeping bag for cold weather. Should they use shiny aluminium material or thick fluffy material? Explain using what you know about heat and insulators.
4	**Roofing Materials** A builder needs a roof that won't get too hot in summer. Should they choose dark metal or light-coloured foam? Explain why, using evidence about how heat moves.
5	**Polar Bear Adaptations** A polar bear has white fur, thick blubber, and large paws. Explain how *each* of these adaptations helps it survive in its icy environment.

💡 **Teacher Answers (guide):** Q1: Energy storage for transformation. Q2: Sun higher at midday = more direct light = shorter shadow. Q3: Thick fluffy = insulator traps heat; aluminium conducts cold in. Q4: Light foam reflects heat and insulates; dark metal absorbs and conducts heat. Q5: White fur = camouflage; blubber = insulation + energy; large paws = weight distribution on ice + swimming.

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Your Science Mission

Now it's your turn to investigate like a real scientist. Choose something around you that makes you curious.

🔍 What do you notice?

Describe what you see, hear or observe around you right now.

❓ What do you wonder?

What question could you ask about it? There are no silly questions in science.

🤔 What do you predict?

What do you think will happen? Commit to a prediction before you test.

📊 What evidence could you collect?

How would you test your idea fairly? What would you measure?

💡 What did you discover?

What does your evidence tell you? Did it match your prediction?

> **Every discovery begins with curiosity.**
> *Your questions matter. Your investigations matter. You are a scientist.*

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